

ADVANCED WATER SAFETY

Differentiated Extension Workbook | Year 9-10 | Victorian Curriculum F-10 Version 2.0

For high-achieving students.
Deeper analysis, original research, statistical modelling,
diving/spinal injury analysis, and policy-level thinking.
VCHPEP143 | VCHPEP144 | VCHPEP145 | VCHPEP148 | VCHPEM162

TEACHER

★ **This is an extension workbook.**

Questions require sophisticated analysis, original research, and evidence-based argument — not description or summary. For each question ask yourself: Am I explaining WHY, not just WHAT? Am I citing primary sources with URLs? Am I acknowledging counter-arguments?

1 Advanced Data Analysis

Q1.1 [5 marks]

Statistical Modelling

Using Victorian drowning data from LSV (lsv.com.au/research), calculate drowning rates per 100,000 for THREE consecutive years. Plot on a line graph, apply a linear trend line, and extrapolate: what would the predicted 2026-27 rate be? What are the limitations of this prediction?

Show all calculations. Discuss at least two factors that could cause actual figures to deviate significantly from your prediction.

Q1.2 [4 marks]

Disaggregation Challenge

Victoria's overall rate is 0.78 per 100,000. Calculate the drowning rate specifically for: (a) Victorian males, and (b) CALD Victorians. Compare these to the overall rate and explain what disaggregating the data reveals that aggregate statistics hide.

For CALD: use 40% of deaths and 28% of population. For males: 46 of 54 deaths, approx 50% of population.

Q1.3 [4 marks]

International Statistical Comparison

Research and compare Victorian drowning rates against TWO other countries using primary sources (WHO, CDC, or national drowning organisations). Write a 150-word analysis explaining what the comparison reveals — and what it does NOT reveal — about Victorian water safety effectiveness.

2 Physiology — Advanced Applications

Q2.1 [4 marks]

Physiological Mechanism Analysis

Explain the specific physiological mechanism that limits swimming ability during the 10-minute cold water window. Your answer must include: peripheral vasoconstriction, neuromuscular dysfunction, and why fitness level does NOT significantly extend this window.

Q2.2 [5 marks]**Evaluating CPR Evidence**

Locate ONE peer-reviewed study or national clinical guideline addressing bystander CPR survival rates in drowning specifically. Record the source, its key finding, and evaluate: (a) how well it supports the 4-minute window claim, (b) its limitations, (c) whether it changes your advice about CPR instruction in schools.

Search: Google Scholar, PubMed, Australian Resuscitation Council (resus.org.au)

3**Structural Analysis — Advanced Social Determinants****Q3.1 [5 marks]****Intersectionality Application**

Apply the intersectionality framework to Victorian drowning risk. Identify a specific profile of a person who faces MULTIPLE intersecting risk factors — explain how each factor amplifies the others rather than simply adding to them.

Reference at least two primary data sources.

Consider: a recently arrived CALD male, aged 19, in a regional area who fishes commercially.

Q3.2 [5 marks]**Health Equity Analysis**

Distinguish between health equality and health equity. Then argue: should a Victorian Government water safety prevention budget be distributed equally or equitably? Acknowledge and refute the strongest counter-argument:

"Targeting resources to specific groups is discriminatory."

4**Prevention Strategy — Advanced Evaluation****Q4.1 [4 marks]****Evidence Quality Assessment**

Locate the specific research evidence underpinning the claim that "four-sided pool isolation fencing reduces child drowning by approximately 83%." Evaluate: (a) what type of study produced this figure, (b) transferability to contemporary Victoria, (c) what would be needed to update it.

Q4.2 [5 marks]**Cost-Effectiveness Analysis**

Research or estimate the approximate cost of: (a) extending patrolled beach hours by 20% across Victoria, (b) funding CALD community swim programs reaching 10,000 participants/year. Estimate the drowning deaths that might be prevented annually. Which provides better cost-effectiveness? What are the significant limitations of this type of analysis?

This requires synthesis and estimation. Reasoning and evidence quality matter more than exact figures.

5 Original Research & International Policy Analysis

Q5.1 [5 marks]**Primary Source Research**

Locate and read the most recent WHO publication specifically on drowning prevention. Identify ONE recommendation NOT currently reflected in Victoria's approach. Write a policy brief paragraph (150-200 words) arguing for its adoption, addressed to the Victorian Minister for Sport.

WHO Drowning Prevention: A Guide to Implementing Recommendations — search who.int publications.

6 Diving, Spinal Injuries & High-Risk Behaviour — Extension Analysis

Extension focus:

This section requires primary source research, original policy argument, and intersectionality analysis applied to a new content domain. Cite all sources with full URLs. A-standard reasoning throughout.

Q6.1 [5 marks]**Primary Source Research: Spinal Injury Data**

Locate the AIHW report on aquatic injury OR the Royal Life Saving Australia spinal injury data (aihw.gov.au or royallifesaving.com.au). Identify ONE specific finding NOT mentioned in the standard workbook. Explain: (a) what the finding is, (b) what it reveals about the scope or distribution of the problem, and (c) how it would strengthen or complicate a prevention policy argument.

Search: AIHW aquatic injury Australia OR RLSSA spinal injury diving. Retrieve a primary document, not a news summary.

Q6.2 [5 marks]**Intersectionality: Compound Risk Analysis**

The data shows 95% of aquatic spinal injury patients are male, the highest incidence group is 10-14 year olds, and inland waterways account for the majority of serious incidents. Apply the intersectionality framework to construct a compound risk profile. Identify a specific type of person who faces intersecting, mutually amplifying risk factors — and argue why addressing any single factor in isolation will be insufficient. Reference at least two statistics and acknowledge one counter-argument.

Intersectionality: risk factors multiply each other — they do not simply add. A rural teenage male faces compounded risk from masculinity norms + limited swim access + no lifeguard infrastructure + alcohol availability + delayed emergency response. Explain the amplification.

Q6.3 [4 marks]**False Confidence as a Structural Problem**

The three Australian case studies (Terrigal Wharf NSW 2025, Torquay Front Beach Vic 2025, Bawley Beach NSW 2024) all involve false confidence — either "it looked safe" or "I have done it before." Argue that false confidence near water is NOT primarily an individual attitude problem but a structural communications failure. What should water safety organisations, local councils, and schools be doing differently — and why are current approaches insufficient? Cite at least one primary source.

Think about: the difference between warning signs (which address knowledge) and behaviour change infrastructure (which addresses the decision-making environment).

Q6.4 [5 marks]**Policy Argument: Diving Prevention vs Other Priorities**

Diving and shallow-water spinal injuries account for 10% of spinal cord injuries and 20% of quadriplegia cases in Australia — but receive significantly less public health policy attention than drowning prevention. Construct a fully evidenced argument for why the Victorian Government should allocate a dedicated budget line to diving injury prevention separate from drowning prevention spend. Your argument must: cite at least two statistics, address the strongest counter-argument ("drowning kills more people"), include a specific costed intervention proposal, and name a success metric.

Consider: cost of lifetime quadriplegia care vs cost of prevention; age distribution of victims (10-14 highest); permanence of spinal injury; WHO cost-effectiveness frameworks from Section 4.

Q6.5 [4 marks]**Evaluating the "Feet First" Message**

Australian water safety organisations have promoted "Feet first, first time" for decades. Evaluate its effectiveness using the evidence hierarchy from Section 4. Is this message addressing the right behaviour? Does the evidence support it as the primary prevention strategy for this risk group? What would a stronger evidence base look like, and what alternative or complementary strategy would you argue for?

Apply the same evaluative framework as pool fencing and patrol hours: evidence type, causal mechanism, population reached, what it does not address.

7**Extension Self-Assessment****— Extension Self-Assessment**

- ☐ I have cited at least 5 primary sources with full URLs
- ☐ I have acknowledged and directly refuted at least one counter-argument
- ☐ My statistical calculations show working and include a reasonableness check
- ☐ I have proposed specific, measurable solutions — not just identified problems
- ☐ My writing is addressed to a specific, named audience — not written generically